[Soal] Geometry Everyday Project

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July 24, 2025

Daftar Isi

§1 Soal

§1.1 Korea MO 2022, Problem 1

Let ABC be an acute triangle with circumcenter O, and let D, E, and F be the feet of altitudes from A, B, and C to sides BC, CA, and AB, respectively. Denote by P the intersection of the tangents to the circumcircle of ABC at B and C. The line through P perpendicular to EF meets AD at Q, and let R be the foot of the perpendicular from A to EF. Prove that DR and OQ are parallel.

§1.2 Japan MO 2022, Problem 3

In an isosceles triangle ABC, $AB = AC$. Take a point O inside the triangle (not on the boundary). Draw a circle ω with center O and passing through C. Suppose $\omega \cap BC = \{C, D\}$, $\omega \cap AC = \{C, E\}$. Denote the circumcircle of $\triangle AEO$ as Γ. And suppose $\Gamma \cap \omega = \{E, F\}$. Prove that the circumcenter of $\triangle BDF$ is on Γ.

§1.3 IMO 2014, Problem 4

Let P and Q be on segment BC of an acute triangle ABC such that $\angle PAB = \angle BCA$ and $\angle CAQ = \angle ABC$. Let M and N be the points on AP and AQ, respectively, such that P is the midpoint of AM and Q is the midpoint of AN. Prove that the intersection of BM and CN is on the circumference of triangle ABC.

§1.4 USA January TST for IMO 2016, Problem 3

Let ABC be an acute scalene triangle and let P be a point in its interior. Let A_1, B_1, C_1 be projections of P onto triangle sides BC, CA, AB, respectively. Find the locus of points P such that AA_1, BB_1, CC_1 are concurrent and $\angle PAB + \angle PBC + \angle PCA = 90°$.

§1.5 Singapore MO Open 2022

For $\triangle ABC$ and its circumcircle ω, draw the tangents at B, C to ω meeting at D. Let the line AD meet the circle with center D and radius DB at E inside $\triangle ABC$. Let F be the point on the extension of EB and G be the point on the segment EC such that $\angle AFB = \angle AGE = \angle A$. Prove that the tangent at A to the circumcircle of $\triangle AFG$ is parallel to BC.

§1.6 Iran-Ukraine Camp 2021 Exam I P-1

Incircle of the triangle $\triangle ABC$ is tangent to sides BC, AC, AB at points D, E, F respectively. Let I be the incentre of triangle and K be the intersection point of AI, BC. Circumcircle w of the triangle $\triangle IDK$ intersects IB and IC at points P and Q. Prove that lines EF, PQ and the tangent to w from D are concurrent.

§1.7 USAMO 2015, Problem 2

Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.

§1.8 All-Russian 2022 Grade 11 Problem 3

An acute-angled triangle ABC is fixed on a plane with largest side BC. Let PQ be an arbitrary diameter of its circumscribed circle, and the point P lies on the smaller arc AB, and the point Q is on the smaller arc AC. Points X, Y, Z are feet of perpendiculars dropped from point P to the line AB, from point Q to the line AC and from point A to line PQ. Prove that the center of the circumscribed circle of triangle XYZ lies on a fixed circle.

Terjemahan: Diberikan sebuah segitiga lancip ABC yang tetap di sebuah bidang dengan sisi terpanjangnya adalah BC. Misalkan PQ adalah sebuah diameter sembarang dari lingkaran luar segitiga ABC, dengan titik P berada di busur AB yang lebih kecil dan titik Q berada di busur AC yang lebih kecil. Titik X, Y, Z berturut-turut adalah kaki tinggi titik P ke garis AB, titik Q ke garis AC, dan titik A ke garis PQ. Buktikan bahwa pusat lingkaran luar segitiga XYZ berada pada sebuah lingkaran tetap.

§1.9 All-Russian 2022 Grade 9 Problem 2

Given triangle ABC with incenter I and A-excenter J. Circle ω_b centered at point O_b passes through point B and is tangent to line CI at point I. Circle ω_c with center O_c passes through point C and touches line BI at point I. Let O_bO_c and IJ intersect at point K. Find the ratio IK/KJ.

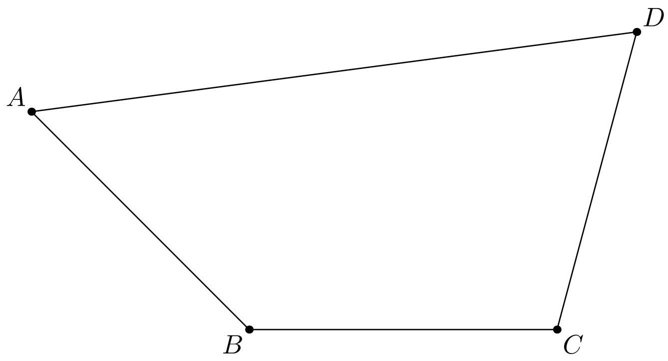
§1.10 CGMO 2022, Problem 3

In triangle $ABC, AB > AC, I$ is the incenter, AM is the midline. The line crosses I and is perpendicular to BC intersect AM at point L, and the symmetry of I with respect to point A is J

Prove that: $\angle ABJ = \angle LBI$.

§1.11 IMSO 2021

In the diagram below, $AB = BC = CD$, $\angle ABC = 135°$ and $\angle BCD = 105°$. What is the measure, in degrees, of $\angle ADC$?



§1.12 Philippine MO 2022, Problem 6

In $\triangle ABC$, let D be the point on side BC such that $AB + BD = DC + CA$. The line AD intersects the circumcircle of $\triangle ABC$ again at point $X \neq A$. Prove that one of the common tangents of the circumcircles of $\triangle BDX$ and $\triangle CDX$ is parallel to BC.

§1.13 Finnish National High School Competition 2019, Problem 3

Let $ABCD$ be a cyclic quadrilateral whose side AB is at the same time the diameter of the circle. The lines AC and BD intersect at point E and the extensions of lines AD and BC intersect at point F. Segment EF intersects the circle at G and the extension of segment EF intersects AB at H. Show that if G is the midpoint of FH, then E is the midpoint of GH.

§1.14 Finnish National High School Competition 2001, Problem 1

In the right triangle ABC, CF is the altitude based on the hypotenuse AB. The circle centered at B and passing through F and the circle with centre A and the same radius intersect at a point of CB. Determine the ratio $FB : BC$.

§1.15 Iberoamerican 2022, Problem 5

Let ABC be an acute triangle with circumcircle Γ. Let P and Q be points in the half plane defined by BC containing A, such that BP and CQ are tangents to Γ and $PB = BC = CQ$. Let K and L

be points on the external bisector of the angle $\angle CAB$, such that $BK = BA, CL = CA$. Let M be the intersection point of the lines PK and QL. Prove that $MK = ML$.

§1.16 Random Problem

Diberikan persegi panjang $ABCD$. Titik P adalah perpotongan garis BC dengan garis yang melalui A dan tegak lurus AC. Titik Q pada segmen CD. Garis PQ memotong garis AD di R. Garis BR memotong garis AQ di X. Buktikan bahwa jika titik Q bergerak sepanjang segmen CD, maka besar $\angle BXC$ konstan.

(Authored by Wildan Bagus Wicaksono)

§1.17 USAJMO 2023, Problem 2

In an acute triangle ABC, let M be the midpoint of $\overline{BC}$. Let P be the foot of the perpendicular from C to AM. Suppose that the circumcircle of triangle ABP intersects line BC at two distinct points B and Q. Let N be the midpoint of $\overline{AQ}$. Prove that $NB = NC$.

§1.18 Korea 2023, Problem 1

In a triangle ABC ($\overline{AB} < \overline{AC}$), points $D(\neq A, B)$ and $E(\neq A, C)$ lies on side AB and AC respectively. Point P satisfies $\overline{PB} = \overline{PD}, \overline{PC} = \overline{PE}$. $X(\neq A, C)$ is on the arc AC of the circumcircle of triangle ABC not including B. Let $Y(\neq A)$ be the intersection of circumcircle of triangle ADE and line XA. Prove that $\overline{PX} = \overline{PY}$.

§1.19 IMO 2023, Problem 2

Let ABC be an acute-angled triangle with $AB < AC$. Let Ω be the circumcircle of ABC. Let S be the midpoint of the arc CB of Ω containing A. The perpendicular from A to BC meets BS at D and meets Ω again at $E \neq A$. The line through D parallel to BC meets line BE at L. Denote the circumcircle of triangle BDL by ω. Let ω meet Ω again at $P \neq B$. Prove that the line tangent to ω at P meets line BS on the internal angle bisector of $\angle BAC$.

§1.20 Japan 2023, Problem 2

In an acute triangle ABC, let D, E, F be the midpoints of BC, CA, AB respectively. Let X, Y be the feet of altitude from D to AB, AC respectively. The line through F parallel to XY meets line DY at P. Prove that $AD \perp EP$.

§1.21 Random Problem 2

Diberikan segitiga ABC dengan besar sudut $\angle ABC = 60^\circ$ dan $\angle BCA = 100^\circ$. Pada sisi AB dan AC berturut-turut terdapat titik D dan E sehingga $\angle EDC = 2\angle BCD = 2\angle CAB$. Tentukan besar sudut $\angle BED$.

§1.22 LMNAS SMP 33 Penyisihan nomor 16

Diberikan $\triangle ABC$ dengan $AB = AC$. Lingkaran dalam $\triangle ABC$ menyinggung sisi AB dan BC berturut-turut di P dan Q. Garis CP memotong lingkaran dalam $\triangle ABC$ di titik P dan R. Jika $PQ = 4$ dan $PR = 5$, panjang QR adalah ...
a.$\frac{1}{2}\sqrt{14}$ b.2 c.$\frac{3}{5}\sqrt{14}$ d.$\frac{4}{5}\sqrt{14}$ e.3

§1.23 LMNAS SMP 33 Penyisihan nomor 3 isian singkat

Diberikan segitiga tumpul ABC dengan $AB = BC$. Titik D berada di dalam segitiga tersebut sedemikian sehingga $AD = BD$, $\angle ADB = 140^\circ$, dan $\angle ADC = 150^\circ$. Besar sudut ACD dalam satuan $^\circ$ (derajat) adalah . . .

§1.24 LMNAS SMP 31 Semifinal nomor 17

Diberikan segitiga lancip ABC. Garis $\ell_1, \ell_2 \perp AB$ berturut-turut pada A dan B. Misalkan M titik tengah AB. Garis melalui M tegak lurus AC dan BC memotong ℓ_1 dan ℓ_2 berturut-turut pada E dan F. Misalkan D titik potong dari EF dan MC. Jika $EF = \frac{64}{3}$, $FM = 16$, dan $BD = 12$. Luas segitiga AMF dapat dinyatakan dalam $a\sqrt{b}$ dimana a dan b bilangan asli dan b tidak dapat dibagi oleh bilangan kuadrat sempurna selain 1. Nilai dari $a + b$ adalah ...

§1.25 Pelatda Pra OSK SMA 2024 DKI Jakarta

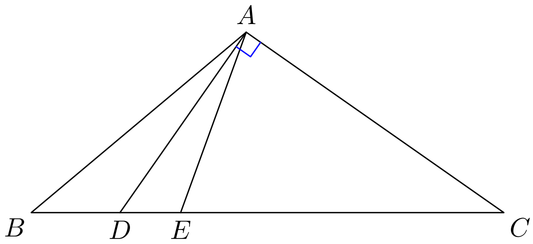
Pada segitiga ABC, titik D adalah kaki garis tinggi dari C. Titik E dan F pada AC dan BC, berturut-turut $AE = AD$ dan $BF = BD$. Titik S adalah refleksi C pada titik pusat lingkaran luar $\triangle ABC$. Buktikan bahwa $SE = SF$.

§1.26 IMO 2024, Problem 4

Let ABC be a triangle with $AB < AC < BC$. Let the incenter and incircle of triangle ABC be I and ω, respectively. Let X be the point on line BC different from C such that the line through X parallel to AC is tangent to ω. Similarly, let Y be the point on line BC different from B such that the line through Y parallel to AB is tangent to ω. Let AI intersect the circumcircle of triangle ABC at $P \neq A$. Let K and L be the midpoints of AC and AB, respectively. Prove that $\angle KIL + \angle YPX = 180°$. (Proposed by Dominik Burek, Poland)

§1.27 Random Problem 3

In the diagram below, AD is perpendicular to AC and $\angle BAD = \angle DAE = 15$. If $AB + AE = BC$, find $\angle ABC$ in degrees.



§1.28 KTOM Januari 2025, Nomor 3 Esai

Diberikan persegi $ABCD$ dan titik E pada ruas CD. Misalkan F adalah titik pada ruas AE sehingga BF tegak lurus AE. Dengan cara serupa, misalkan G adalah titik pada ruas EB sehingga AG tegak lurus EB. Misalkan garis DF dan CG berpotongan di H. Buktikan bahwa

$$\angle DFC + \angle DGC = 90° + \angle AHD + \angle BHC.$$

§1.29 OSN SMP 2016, Nomor 3, Hari Pertama

Pada segitiga ABC, titik P dan Q berada pada sisi BC sehingga panjang BP sama dengan CQ. $\angle BAP = \angle CAQ$ dan $\angle APB$ lancip. Apakah segitiga ABC sama kaki? Tulis alasan Anda.

§1.30 Random Problem 4

Diberikan segitiga ABC dan lingkaran luar ω dari segitiga tersebut. Ditarik sembarang garis yang sejajar BC, yang memotong sisi AB dan AC secara berurutan di titik X dan Y. Misalkan M adalah titik tengah dari busur BC pada lingkaran ω yang tidak memuat titik A. Garis MX dan MY memotong lingkaran ω kembali di titik S dan T secara berurutan. Misalkan ST dan XY berpotongan di titik P. Tunjukkan bahwa garis AP menyinggung lingkaran ω.

§1.31 A lemma from Berkeley Math Tournament 2024

The incircle of scalene triangle $\triangle ABC$ is tangent to $\overline{BC}, \overline{AC}$, and $\overline{AB}$ at points D, E, and F, respectively. The line EF intersects line BC at P and line AD at Q. The circumcircle of $\triangle AEF$

intersects line AP again at point $R \neq A$. If I is the incenter of $\triangle ABC$, prove that I, Q, R are collinear.

§1.32 Japan MO 2025, Problem 2

Let ABC be an acute triangle and let O be its circumcenter. Let O_1 and O_2 be the circumcenters of triangles ABO and ACO, respectively. The circumcircle of triangle AO_1O_2 intersects the side BC (excluding its endpoints) at two distinct points P and Q, such that the four points B, P, Q, C are collinear in this order. Let O_3 be the circumcenter of triangle OPQ. Prove that the three points A, O, O_3 are collinear.

§1.33 OSP SMA 2019, Essay, Problem 5

Diberikan segitiga ABC, dengan $AC > BC$, dan lingkaran luarnya yang berpusat di O. Misalkan M adalah titik pada lingkaran luar segitiga ABC sehingga CM adalah garis bagi $\angle ACB$. Misalkan Γ adalah lingkaran berdiameter CM. Garis bagi $\angle BOC$ dan garis bagi $\angle AOC$ memotong Γ berturut-turut di P dan Q. Jika K adalah titik tengah CM, buktikan bahwa P, Q, O, K terletak pada satu lingkaran.